

gei-17 (Q94361): The PIAS-family SUMO E3 Ligase of *Caenorhabditis elegans*

Gene/Protein Identity Verification

This report concerns **gei-17** (UniProt **Q94361**; WormBase/ORF **W10D5.3**), the *Caenorhabditis elegans* gene encoding an **E3 SUMO-protein ligase** of the **PIAS family** (EC 2.3.2.-). Identity was verified against the UniProt-provided target throughout:

- **Protein family match:** All primary literature describes GEI-17 as the *C. elegans* **PIAS** ortholog ("PIAS(GEI-17)"), matching the UniProt "Belongs to the PIAS family" designation.
- **Domain match:** UniProt lists the diagnostic PIAS domains **PINIT** (PF14324/IPR023321, residues ~203–367) and the **SP-RING / MIZ-type zinc finger** (Znf_MIZ IPR004181; RING/FYVE/PHD fold IPR013083, residues ~400–485). These are the catalytic hallmarks of the PIAS/Siz1 SUMO-ligase class.
- **Organism match:** Every cited study is in *C. elegans*.
- **Name origin:** "GEI" denotes a **Gex-3 (GEX-3)-interacting protein**; "gei-17" was named as a Gex-3-interacting protein, but its molecular identity is the nematode PIAS SUMO E3 ligase.

The gene symbol is **unambiguous** for this protein, and a well-developed body of primary *C. elegans* literature supports the functional annotation below. There is no cross-organism symbol confusion for "gei-17."

Summary

gei-17 encodes the single PIAS-family SUMO E3 ligase of *C. elegans*, a nuclear, zinc-dependent protein transferase (EC 2.3.2.-) that catalyzes the covalent attachment of the small ubiquitin-like modifier **SMO-1** (the worm SUMO/SMT3 ortholog) onto lysine residues of specific substrate proteins. GEI-17 works within the canonical SUMO cascade: the SUMO-activating E1, the single E2 conjugating enzyme **UBC-9**, and GEI-17 as the substrate-selecting E3. Mechanistically, its **SP-RING** domain recruits the charged E2~SUMO (UBC-9~SMO-1) thioester while its **PINIT** domain supports substrate/SUMO engagement — a mechanism established

experimentally for the orthologous yeast PIAS ligase Ull1/Siz1 and inferred for GEI-17 by conserved domain architecture. In vitro, GEI-17 both SUMOylates substrates and undergoes extensive **auto-SUMOylation**.

GEI-17 is a **nuclear, chromatin-associated enzyme** that acts as a hub for regulated SUMOylation in three major arenas. **(1) Genome maintenance and DNA-damage tolerance:** GEI-17 was first discovered as the E3 that SUMOylates **MUS-101/TopBP1**, and it enables replication of damaged chromosomes by SUMOylating and stabilizing the translesion polymerase **POLH-1 (pol η)**, protecting it from CRL4-Cdt2-mediated proteolysis until translesion synthesis is complete. Together with *polh-1* and *rad-2*, GEI-17 **silences the ATL-1/CHK-1 (ATR-Chk1) DNA-damage checkpoint** in early embryos by suppressing replication-fork stalling. **(2) Chromosome segregation:** Through dynamic, spatially organized SUMOylation, GEI-17 drives both mitotic chromosome alignment and acentrosomal oocyte-meiotic segregation, building SUMO–SIM scaffolds and modifying the chromokinesin **KLP-19**, the kinase **BUB-1**, and the CLASP ortholog **CLS-2**. **(3) Transcription and nuclear organization:** GEI-17 SUMOylates specific transcription factors (the ETS factor **LIN-1** at K169; the T-box factor **TBX-2**), is required for **perinuclear telomere anchoring**, and regulates **piRNA transcription condensate** formation.

The unifying theme is that GEI-17 is a **nuclear SUMO-writing enzyme** that confers regulatory information — stability, localization, protein–protein scaffolding, and activity changes — onto chromatin-associated substrates during DNA replication/repair, chromosome segregation, and transcription. Its function is executed in the nucleus, on chromatin, at replication forks, and on the chromosome–spindle apparatus.

Key Findings

Finding 1 — GEI-17 is the sole PIAS-family SUMO E3 ligase in *C. elegans*, acting with E2 UBC-9 and SUMO SMO-1

GEI-17 is annotated in UniProt (Q94361) as "E3 SUMO-protein ligase gei-17; EC 2.3.2.-; Belongs to the PIAS family," a ~780-residue (~88 kDa) protein carrying the diagnostic **PINIT** domain (PF14324/IPR023321) and the **SP-RING / MIZ-type zinc finger** (IPR004181; a RING/FYVE/PHD-type zinc-binding fold, IPR013083). Multiple genetic and biochemical studies establish GEI-17 as the nematode PIAS ortholog that functions in the SUMO conjugation pathway alongside the single E2 conjugating enzyme **UBC-9** and the SUMO ortholog **SMO-1** (SMT3).

The direct biochemical evidence that GEI-17 is a catalytically active E3 comes from in vitro reconstitution. Pelisch and colleagues showed that *"In vitro analysis revealed that KLP-19 is efficiently sumoylated in a GEI-17-dependent manner; while GEI-17 undergoes extensive auto-sumoylation"* (PMID: 27939944). This demonstrates both substrate SUMOylation activity and the characteristic PIAS auto-SUMOylation. Genetically, GEI-17 is paired with the E2 in vivo: *"Accumulation of SUMO conjugates on the metaphase plate and proper chromosome alignment depend on the SUMO E2 conjugating enzyme UBC-9 and SUMO E3 ligase PIAS(GEI-17)"* (PMID: 25475837). Together these establish GEI-17's core molecular identity: a PIAS-type SUMO E3 ligase catalyzing the transfer of SUMO-1 from UBC-9 to substrate lysines.

Finding 2 — GEI-17 enables replication of damaged chromosomes by SUMOylating and stabilizing translesion polymerase POLH-1 (pol η)

A defining functional role of GEI-17 is in **DNA-damage tolerance**. Kim & Michael (2008) demonstrated that *"Both the POLH-1 (pol eta) translesion synthesis (TLS) DNA polymerase and the GEI-17 SUMO E3 ligase are essential for the efficient replication of damaged chromosomes in Caenorhabditis elegans embryos"* (PMID: 19111656). Upon DNA damage, POLH-1 is normally targeted for destruction by the **CRL4-Cdt2** (Cul4-Ddb1-Cdt2) ubiquitin ligase. GEI-17 counteracts this: *"GEI-17 protects POLH-1 from CRL4-Cdt2-mediated destruction until after it has performed its function in TLS, and this is likely via SUMOylation of POLH-1"* (PMID: 19111656).

This finding is mechanistically precise and illustrates the "regulatory information" model of SUMOylation: GEI-17 does not act as a protease or a kinase, but writes a SUMO mark on POLH-1 that **antagonizes its ubiquitin-dependent proteolysis**, thereby timing the availability of the TLS polymerase to the window when it is needed to bypass lesions. The consequence of this activity — preventing replication-fork stalling on damaged templates — links directly to checkpoint silencing (Finding 5). This role is captured in the curated GO terms "positive regulation of error-prone translesion synthesis" (GO:1904333, IMP) and "negative regulation of protein catabolic process" (GO:0042177, IMP).

Finding 3 — GEI-17 drives mitotic and meiotic chromosome segregation through dynamic, spatially organized SUMOylation

GEI-17 is central to chromosome segregation in both mitotic and meiotic divisions. In **mitosis**, chromatin-associated SUMO conjugates rise at metaphase and fall at anaphase; metaphase-plate SUMO accumulation and proper chromosome alignment require UBC-9 and GEI-17 (PMID: 25475837). The system is dynamic in both directions — deconjugation by the SUMO

protease ULP-4 is needed for Aurora-B (AIR-2) relocation and cell-cycle progression — underscoring that it is the cycle of SUMOylation/deSUMOylation, written by GEI-17 and erased by proteases, that organizes mitotic chromosome behavior.

In **oocyte meiosis**, which in *C. elegans* occurs on an **acentrosomal spindle**, GEI-17 assembles the machinery for chromosome congression. Pelisch et al. (2017) showed that "*SUMO E3 ligase GEI-17/PIAS is required for KLP-19 recruitment to the RC [ring complex], and proteomic analysis identified KLP-19 as a SUMO substrate in vivo*" (PMID: 27939944). The mechanism is a **SUMO–SIM (SUMO–interaction motif) protein network**: "*GEI-17 and another RC component, the kinase BUB-1, contain functional SUMO interaction motifs (SIMs), allowing them to recruit SUMO modified proteins, including KLP-19, into the RC*" (PMID: 27939944). GEI-17 therefore builds a self-organizing scaffold: it SUMOylates substrates and, via its own SIM and those of partner proteins, cross-links SUMOylated components into the inter-homolog ring complex that drives chromosome congression and segregation.

Finding 4 — GEI-17 SUMOylates specific transcription factors and regulates telomere anchoring and piRNA transcription

Beyond replication and segregation, GEI-17 modifies the activity and localization of **nuclear transcription and chromatin factors**. During **vulval development**, tissue-specific auxin-induced degradation (AID) of GEI-17/SMO-1 disrupts anchor-cell positioning, basement-membrane breaching, vulval precursor cell fate, and morphogenesis; crucially, "*sumoylation of the ETS transcription factor LIN-1 at K169 is necessary for the proper contraction of the ventral vulA toroids*" (PMID: 35666766). This identifies a specific substrate (LIN-1), a specific modified residue (K169), and a defined morphogenetic consequence.

GEI-17 is also linked to the T-box transcription factor **TBX-2**: "*TBX-2 contains 2 consensus sumoylation sites, and it interacts in a yeast two-hybrid assay with the UBC-9 and GEI-17 components of the C. elegans SUMO-conjugating pathway*" (PMID: 16701625), a function required for ABA-derived pharyngeal muscle development.

In **nuclear architecture**, perinuclear telomere anchoring in embryos requires GEI-17: "*Telomere position in early embryos required the NE protein SUN-1, the single-strand binding protein POT-1, and the small ubiquitin-like modifier (SUMO) ligase GEI-17*" (PMID: 24297748). Finally, GEI-17 regulates **piRNA transcription**: "*we searched for factors that regulate piRNA transcription and isolated the SUMO E3 ligase GEI-17 as inhibiting and the SUMO protease TOFU-3 as promoting piRNA transcription foci formation, thereby regulating piRNA production*"

([PMID: 40316696](#)). Here GEI-17 opposes the formation of transcriptional condensates, with the SUMO protease TOFU-3 acting in the opposite direction — again illustrating the writer/eraser balance that tunes GEI-17's outputs.

Finding 5 — GEI-17 (with POLH-1 and rad-2) silences the ATL-1/CHK-1 DNA-damage checkpoint in embryos

A striking developmental role of GEI-17 is in **checkpoint silencing**. In early *C. elegans* embryos, the DNA-damage checkpoint is actively suppressed so that the rapid, scheduled divisions of the P (germline) lineage can proceed. Holway et al. (2006) found that "*the checkpoint response to DNA damage is actively silenced in embryos but not in the germ line,*" and that "*Silencing requires rad-2, gei-17, and the polh-1 translesion DNA polymerase, which suppress replication fork stalling and thereby eliminate the checkpoint-activating signal*" ([PMID: 16549501](#)).

This places GEI-17 upstream in a mechanistically coherent pathway: by stabilizing POLH-1 (Finding 2), GEI-17 ensures efficient translesion synthesis, which prevents replication forks from stalling on damaged templates. Because stalled forks generate the single-stranded DNA/RPA signal that activates ATL-1 (ATR) and downstream CHK-1 (Chk1), suppressing fork stalling **eliminates the checkpoint-activating signal** at its source. A parallel silencing mechanism operates through the SMK-1/PPH-4.1 phosphatase, which recruits protein phosphatase 4 to replicating chromatin to silence the CHK-1 response ([PMID: 17908915](#)); the GEI-17/POLH-1 pathway and the SMK-1/PPH-4.1 pathway together keep the embryonic ATL-1–CHK-1 axis quiet. This is reflected in the GO annotation "negative regulation of mitotic DNA damage checkpoint" (GO:1904290, IMP).

Finding 6 — Domain-based mechanistic inference: PINIT and SP-RING domains execute SUMO transfer

GEI-17's catalytic mechanism can be inferred with high confidence from its conserved domain architecture and from direct experiments on orthologous PIAS/Siz-type ligases. Takahashi & Kikuchi (2005), studying the yeast PIAS-type ligase Ull1/Siz1, showed that "*A novel conserved N-terminal domain, called PINIT, as well as the RING-like domain (SP-RING) were required for the SUMO ligase activity in the in vitro conjugation system and for interaction with Smt3 in an in vitro binding assay*" ([PMID: 16109721](#)). They further defined the catalytic logic of this enzyme class: "*E3 (SUMO ligase) functions as an adaptor between E2 and each substrate*" ([PMID: 16109721](#)).

Because GEI-17 carries both diagnostic domains — **PINIT** (residues ~203–367) and the **SP-RING zinc finger** (residues ~400–485) — the inference is direct: the SP-RING domain recruits and positions the charged E2~SUMO (UBC-9~SMO-1) thioester for catalysis, while the PINIT domain supports substrate and SUMO engagement. In the orthologous enzyme, the N-terminal SAP motif governs DNA binding and nuclear localization rather than catalysis, consistent with GEI-17 being a chromatin-associated enzyme whose catalytic core resides in the PINIT + SP-RING module.

Finding 7 — Original characterization: GEI-17 SUMOylates MUS-101/TopBP1; authoritative GO places it on chromatin in the nucleus

GEI-17 was originally characterized by Holway, Hung & Michael (2005) in a systematic RNAi screen for modifiers of *mus-101* (the *C. elegans* Mus101/**TopBP1** ortholog, a replication and DNA-damage factor). Among chromosome I modifiers, *"we have found five chromosome I genes that modify the mus-101 RNAi phenotype, and we go on to show that one of them encodes an E3 SUMO ligase that promotes SUMO modification of MUS-101 in vitro"* (PMID: [15654100](#)). This is the founding identification of *gei-17* as a SUMO E3 ligase and the demonstration of its in vitro activity toward a specific substrate (MUS-101/TopBP1). UniProt Q94361 cites this work as the defining FUNCTION statement ("Functions as an E3-type smo-1 ligase") and assigns the PATHWAY "protein sumoylation."

Curated **Gene Ontology** annotations in WormBase corroborate the nuclear, chromatin-based identity:

GO aspect	Term	Evidence
Cellular component	chromatin (GO:0000785)	IDA
Cellular component	metaphase plate (GO:0070090)	IDA
Cellular component	nucleoplasm	IBA
Molecular function	SUMO ligase activity (GO:0061665)	IDA
Molecular function	RNA Pol II transcription factor binding; transcription coregulator/inhibitor; zinc ion binding	—
Biological process	DNA damage response	IMP
Biological process	positive regulation of error-prone translesion synthesis (GO:1904333)	IMP
Biological process	negative regulation of mitotic DNA damage checkpoint (GO:1904290)	IMP
Biological process	negative regulation of protein catabolic process (GO:0042177)	IMP
Biological process	meiotic metaphase I homologous chromosome alignment	IMP
Biological process	protein sumoylation	IDA

These annotations independently recapitulate every major experimental finding above: SUMO ligase activity, chromatin/metaphase-plate localization, translesion synthesis, checkpoint suppression, protection from proteolysis, and meiotic chromosome alignment.

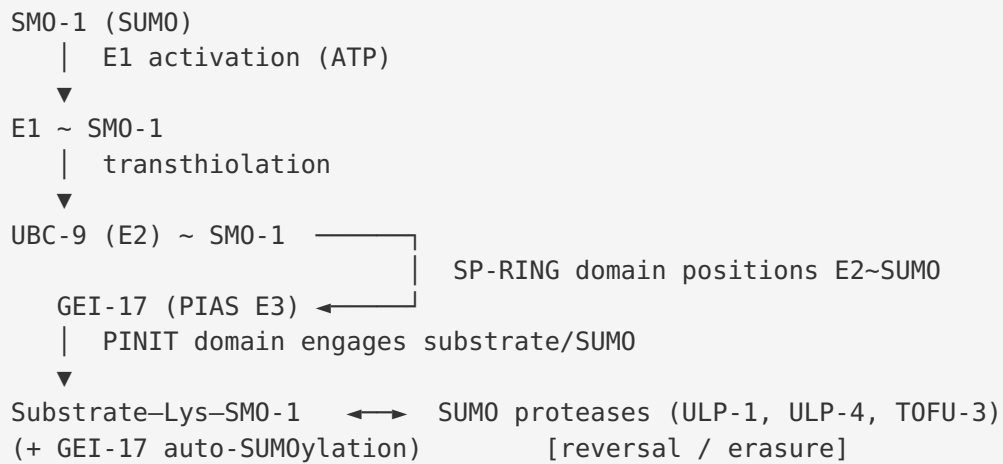
Finding 8 — In oocyte meiosis GEI-17 SUMOylates BUB-1 and regulates CLS-2/CLASP during anaphase I

The meiotic role extends beyond KLP-19 to the kinase BUB-1 and the CLASP ortholog CLS-2. Pelisch et al. (2019) reported that "*SUMO modification of BUB-1 is regulated by the SUMO E3 ligase GEI-17 and the SUMO protease ULP-1. SUMO and GEI-17 are required for BUB-1 localisation between segregating chromosomes during early anaphase I*" (PMID: 31243051). Furthermore, "*CLS-2 is subject to SUMO-mediated regulation; CLS-2 precociously localises in the midbivalent when either SUMO or GEI-17 are depleted*" (PMID: 31243051). CLS-2/CLASP is a microtubule-stabilizing protein that generates poleward force; GEI-17-dependent SUMOylation controls the **timing** of its localization. Thus GEI-17 governs the ordered, timed dynamics of multiple proteins (KLP-19, BUB-1, CLS-2) on the acentrosomal spindle to achieve faithful chromosome segregation during anaphase I.

Mechanistic Model / Interpretation

GEI-17 is best understood as a **nuclear SUMO "writer"** whose single biochemical activity — transferring SUMO-1 from UBC-9 onto substrate lysines — is deployed across several chromatin-based processes. In each case, the SUMO mark confers one of a small set of regulatory outcomes: **protection from proteolysis**, **protein localization/scaffolding via SUMO–SIM interactions**, or **modulation of transcription-factor/condensate activity**. The outputs are shaped not only by GEI-17 but by opposing **SUMO proteases** (ULP-1, ULP-4, TOFU-3), making SUMOylation a dynamic, reversible regulatory layer.

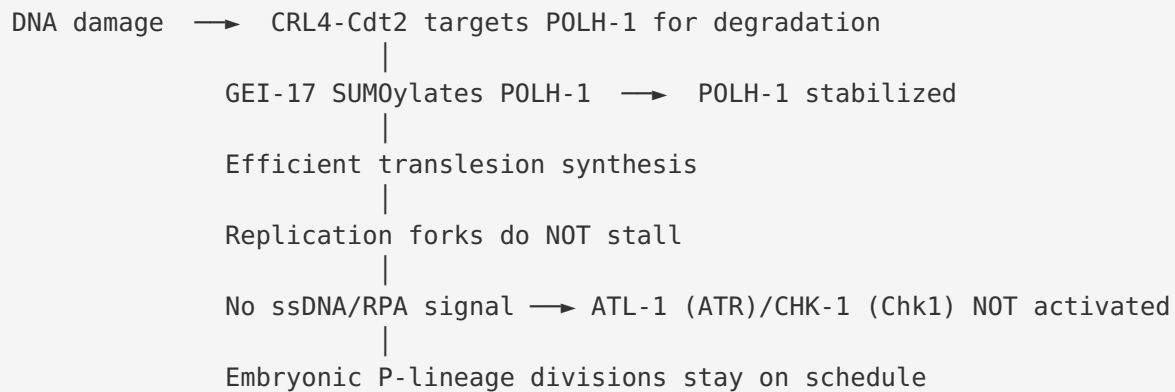
The core enzymatic cascade



Substrate map and functional outputs

Substrate	Process	SUMOylation outcome	Localization	Evidence (PMID)
MUS-101/TopBP1	DNA replication/ damage	SUMO modification (in vitro)	chromatin	15654100
POLH-1 (pol η)	Translesion synthesis	Protection from CRL4-Cdt2 proteolysis; times TLS	replication fork	19111656
KLP-19 (chromokinesin)	Meiotic congression	Recruitment to ring complex via SUMO-SIM	midbivalent/ RC	27939944
BUB-1 (kinase)	Meiotic anaphase I	Localization between segregating chromosomes	spindle midzone	31243051
CLS-2/CLASP	Meiotic force generation	Timing of midbivalent localization	spindle	31243051
LIN-1 (ETS TF)	Vulval morphogenesis	SUMO at K169; vulA toroid contraction	nucleus	35666766
TBX-2 (T-box TF)	Pharyngeal muscle	Consensus SUMO sites; interaction	nucleus	16701625
(telomere complex)	Telomere anchoring	Perinuclear tethering via SUN-1/POT-1	nuclear periphery	24297748

Integrated logic of the DNA-damage / checkpoint module



This single module explains three GO annotations at once: *positive regulation of error-prone translesion synthesis*, *negative regulation of protein catabolic process* (stabilizing POLH-1), and *negative regulation of the mitotic DNA-damage checkpoint*. It is the clearest example of GEI-17's "write a SUMO mark → change a protein's fate → change a cellular decision" logic.

Where GEI-17 works

GEI-17 is a **nuclear** enzyme. Its catalytic actions occur on **chromatin** (SUMO ligase activity, GO:0000785 IDA), at **replication forks** (POLH-1/MUS-101), on the **metaphase plate** and **acentrosomal meiotic spindle/ring complex** (KLP-19, BUB-1, CLS-2), at the **nuclear periphery** (telomere anchoring), and at sites of **piRNA transcription**. It does not have a described extracellular or cytoplasmic catalytic role; its SAP-motif/DNA-binding and nuclear-localization determinants (by orthology to Ull1/Siz1) keep it chromatin-proximal.

Evidence Base

PMID	Title (abbrev.)	Contribution to the annotation
15654100	<i>mus-101 modifier RNAi screen</i>	Founding identification of <i>gei-17</i> as an E3 SUMO ligase; SUMOylates MUS-101/TopBP1 in vitro.
19111656	<i>Regulated proteolysis of pol η</i>	GEI-17 stabilizes POLH-1 via SUMOylation, protecting it from CRL4-Cdt2; enables damaged-chromosome replication.
16549501	<i>Checkpoint silencing in embryos</i>	<i>gei-17</i> (+ <i>polh-1</i> , <i>rad-2</i>) silences the embryonic ATL-1/CHK-1 checkpoint by suppressing fork stalling.
17908915	<i>SMK-1/PPH-4.1 silencing</i>	Parallel checkpoint-silencing pathway; contextualizes GEI-17's role in the embryonic DNA-damage response.
25475837	<i>Dynamic SUMO in mitosis</i>	GEI-17 + UBC-9 required for metaphase-plate SUMO and chromosome alignment; ULP-4 reverses.
27939944	<i>SUMO network in oocyte meiosis</i>	GEI-17 SUMOylates KLP-19; SUMO–SIM scaffold (GEI-17, BUB-1 SIMs) builds the ring complex; auto-SUMOylation shown.
31243051	<i>Sumoylation in meiotic segregation</i>	GEI-17 SUMOylates BUB-1; regulates CLS-2/CLASP timing in anaphase I.
35666766	<i>SUMO in vulval development</i>	GEI-17/SMO-1 sumoylate LIN-1 at K169; diverse developmental SUMO functions.
16701625	<i>TBX-2 and UBC-9</i>	TBX-2 interacts with UBC-9/GEI-17; links GEI-17 to a T-box transcription factor.
24297748	<i>POT-1 telomere anchoring</i>	GEI-17 required for perinuclear telomere anchoring via SUN-1/POT-1.
40316696	<i>piRNA condensate formation</i>	GEI-17 inhibits piRNA transcription foci (TOFU-3 promotes); regulates piRNA production.
16109721	<i>Yeast Ull1/Siz1 domains</i>	Experimental basis for GEI-17 mechanism: PINIT + SP-RING required for SUMO-ligase activity; E3 = E2–substrate adaptor.
27631810	<i>Tools to study SUMO in C. elegans</i>	Methods review establishing the reagents (E3 ligases, SUMO proteases) used to dissect GEI-17 function.
31614335	<i>PIASy regulation of MafA</i>	Mammalian PIAS comparator: PIAS can repress transcription SUMOylation-independently via SIM — relevant to interpreting GEI-17's non-catalytic scaffolding.

Convergence of evidence. The annotation rests on multiple orthogonal lines: (i) **direct in vitro biochemistry** (SUMOylation of MUS-101 and KLP-19; auto-SUMOylation); (ii) **genetics/RNAi/AID** (phenotypes for replication, checkpoint, segregation, development); (iii) **in vivo proteomics** (KLP-19 as a SUMO substrate); (iv) **curated GO (IDA/IMP)**; and (v) **domain-based inference** anchored to experimentally validated orthologs. The DNA-damage/checkpoint and meiotic-segregation stories are each supported by two or more primary papers.

A note on PIAS versatility. The mammalian PIASy–MafA study ([PMID: 31614335](#)) shows that PIAS proteins can also repress transcription through a **SIM-dependent, SUMOylation-independent** mechanism, with the PINIT and SP-RING domains dispensable for that particular repression. This cautions that some of GEI-17's transcriptional/condensate effects (e.g., LIN-1, TBX-2, piRNA foci) may combine catalytic SUMOylation with non-catalytic SIM-mediated scaffolding — a nuance worth testing in the worm.

Limitations and Knowledge Gaps

1. **Direct substrate lysine mapping is sparse.** For several substrates (MUS-101, POLH-1, KLP-19, BUB-1), the modified lysine(s) and the in vivo stoichiometry/dynamics are not fully defined. The best-mapped site is LIN-1 K169. Precise site-level SUMO acceptor maps are largely missing.
2. **Catalytic vs. scaffolding contributions are not separated.** GEI-17 has both SUMO-ligase activity and a SUMO-interaction capacity (SIM). Which outputs require catalysis versus SIM-mediated scaffolding (as seen for mammalian PIASy–MafA) has not been dissected with catalytic-dead (SP-RING mutant) versus SIM-mutant GEI-17 for most processes.
3. **No experimental structure of GEI-17.** The domain-level mechanism is inferred from yeast Ull1/Siz1 and the PIAS family; there is no worm-specific structural data (crystal/cryo-EM/validated AlphaFold analysis) presented here to confirm the E2~SUMO docking geometry.
4. **Redundancy and specificity within the SUMO system.** GEI-17 is described as the sole PIAS-family E3, but the extent to which substrate selection depends on GEI-17 versus the shared E2 UBC-9, versus additional non-PIAS E3-like factors, is incompletely resolved for some substrates.
5. **Opposing proteases.** The erasers (ULP-1, ULP-4, TOFU-3) shape GEI-17's net output, but the full writer/eraser network and its spatial/temporal regulation remain only partially mapped.

6. **Quantitative/structural bioinformatics not performed here.** This report is literature- and annotation-based; no new sequence/structure analysis (e.g., conservation of the SP-RING zinc-coordinating residues, SIM prediction) was independently computed.

Proposed Follow-up Experiments / Actions

1. **Separation-of-function alleles.** Engineer (CRISPR) an SP-RING catalytic-dead GEI-17 and a SIM-mutant GEI-17, and test each across the four arenas (TLS/checkpoint, mitotic alignment, meiotic ring complex, transcription/piRNA) to partition catalytic vs. scaffolding requirements — directly testing the PIASy–MafA-inspired hypothesis.
 2. **In vivo SUMO-site proteomics.** Perform SUMO-remnant immunoaffinity MS in wild-type vs. *gei-17* depletion to map acceptor lysines on POLH-1, MUS-101, KLP-19, BUB-1, and CLS-2, and to define the GEI-17-dependent SUMO proteome.
 3. **Non-degradable/non-SUMOylatable POLH-1.** Test POLH-1 lysine-to-arginine (SUMO-site) and degron mutants to confirm that SUMOylation is the signal protecting POLH-1 from CRL4-Cdt2 and timing TLS.
 4. **Structure determination / validated modeling.** Solve or model (AlphaFold-Multimer, with PAE/pLDDT validation) the GEI-17 SP-RING–UBC-9~SMO-1 complex to confirm the catalytic geometry inferred from Ull1/Siz1.
 5. **Live imaging of SUMO dynamics.** Use the established worm SUMO-reporter toolkit ([PMID: 27631810](#)) to quantify GEI-17-dependent SUMO conjugation/deconjugation in space and time at forks, the metaphase plate, and the meiotic ring complex, with and without the opposing proteases.
 6. **Condensate mechanism for piRNA.** Determine whether GEI-17 inhibits piRNA transcription foci by SUMOylating a specific condensate component (vs. SIM-mediated sequestration), and identify the TOFU-3 target that it counteracts.
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Conclusion

gei-17 (Q94361) encodes the single PIAS-family SUMO E3 ligase of *C. elegans*: a nuclear, zinc-dependent protein transferase (EC 2.3.2.-) whose PINIT and SP-RING/MIZ domains transfer the SUMO ortholog SMO-1 from the E2 UBC-9 onto substrate lysines (with extensive auto-SUMOylation). This single activity underlies three major, well-evidenced functions — (1) DNA-damage tolerance and silencing of the ATL-1/CHK-1 checkpoint via SUMOylation of MUS-101/TopBP1 and stabilization of POLH-1/pol η ; (2) mitotic and acentrosomal-meiotic chromosome segregation via SUMO–SIM scaffolds and modification of KLP-19, BUB-1, and CLS-2/CLASP; and (3) transcriptional/nuclear-organization control (LIN-1 K169, TBX-2, telomere anchoring, piRNA condensates). GEI-17 executes its function **in the nucleus — on chromatin, at replication forks, and on the chromosome–spindle apparatus.**

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